ZB631KL Service Manual

V1.0

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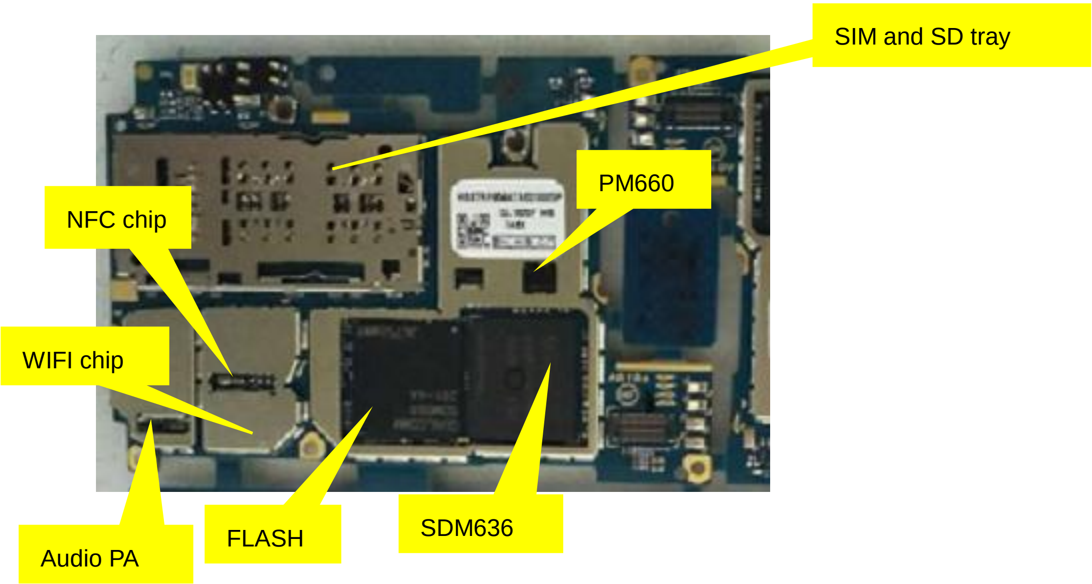
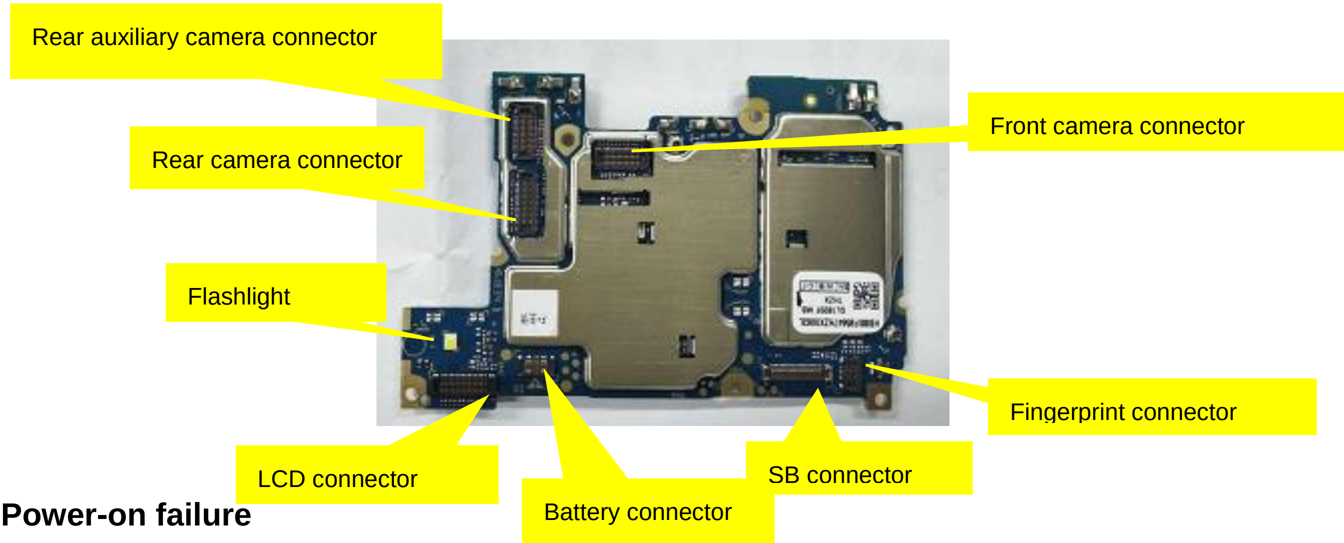
Service Manual

1. MB illustration





2. Power-on failure

Solution:

1 Power on and observe the MB power-on current. If there is no current in the keys, insert a USB and observe whether or not it will power on. If so, test the key module circuit

2 Check the VBAT power supply test point on the MB for dirt. Check the battery connector for body damages or soldering errors.

3 If the power-on current is less than 200MA, troubleshoot the following parts. Download the software again and troubleshoot the crystal circuit components.

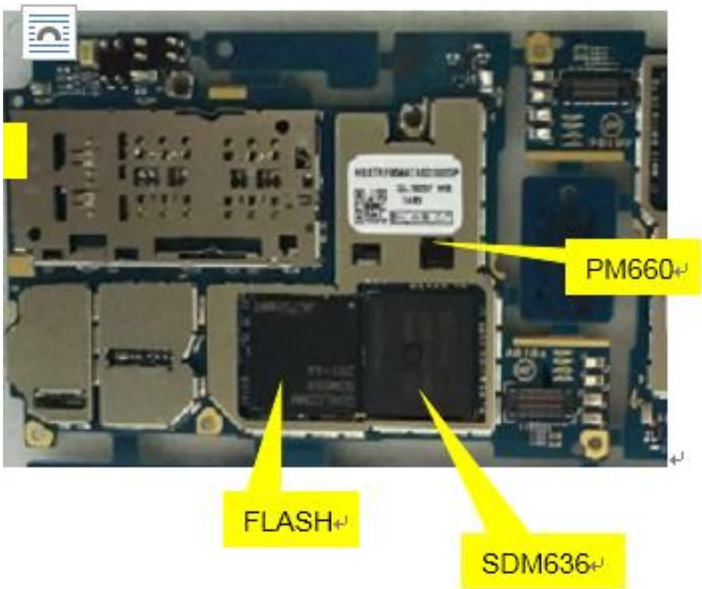
4 For power on current errors above 1A, check the VBAT circuit for short circuits to ground, troubleshoot the ground diode at individual locations;

5 For power-on current errors above 200MA, check for hot point on the MB and troubleshoot;

6 Check main chips with an X-ray for soldering errors, including the SDM636/Flash/PMICTR and DC power chips.

7 Add-solder or re-solder the main chip, validate the soldering conditions;

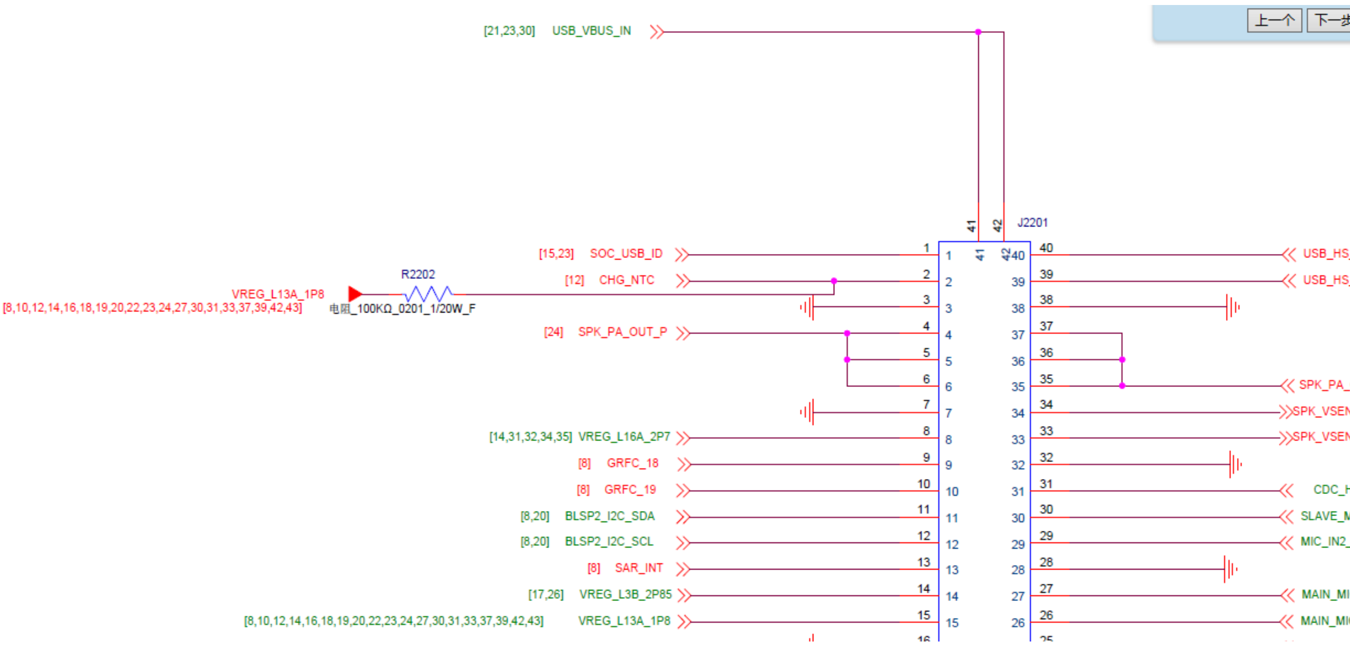
8 Replace SDM636/Flash/PM660/TR IC.



3. Blank screen

Solution:

1. Check the LCD connector for dirt, PIN continuous tin, cold solder joint, PIN collapse, etc. Check the connector for proper connection and re-install it.
2. Re-test after replacing the LCD and FPC.
3. LCD backlight power is provided by U2001. Measure and compare the voltages (LED-K&LED-A)
4. Check R2202 and peripheral zone for soldering and body resistance errors;
5. Check U2001 with an X-ray for soldering errors;



4. Screen display abnormality

Solution:

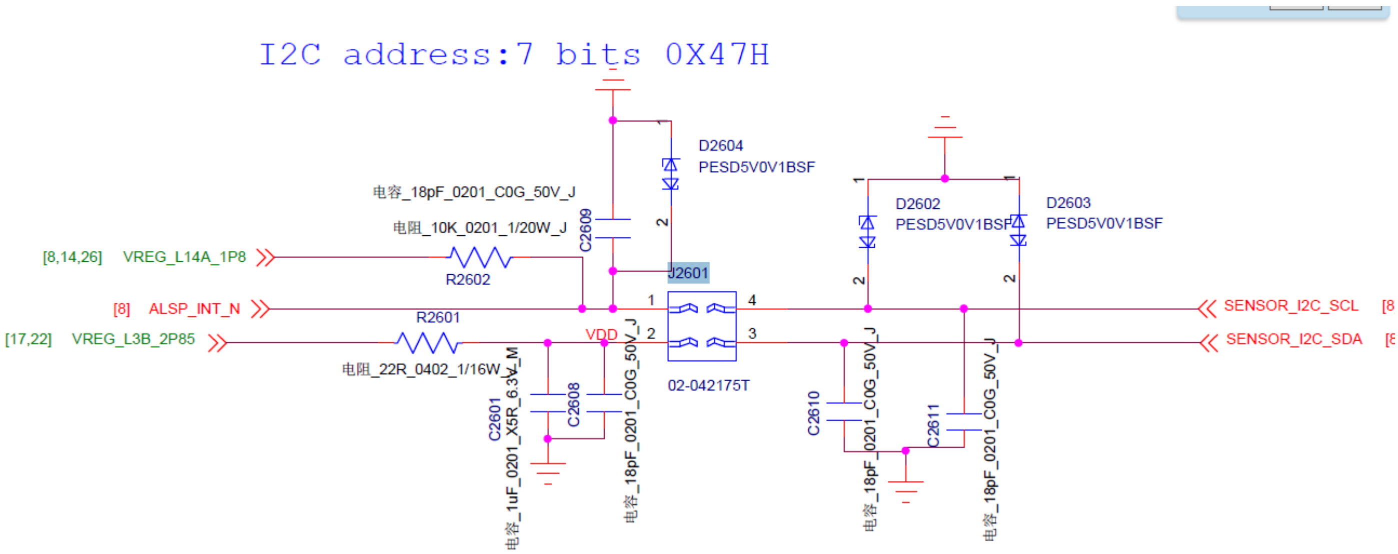
1 Re-install the LCD display screen or replace it before powering on the phone and check again;
2 Check the LCD connector for dirt, PIN continuous tin, cold solder joint, PIN collapse, etc.;
3 Check the LCD filter (T2001, T2002, T2003, T2004, T2005) for soldering errors or damages;
4 Check the LCD peripheral components. Compare them with those on the normal board to check for missing parts. Test supply voltage and check whether or not it is normal (VPH-PWR);
5 Check the CPU for cold solder joint. Check IC with an X-ray, if available, for solder balls;
6 Add-solder or re-solder the CPU or replace it;

5. Touch-screen failure

Solution:

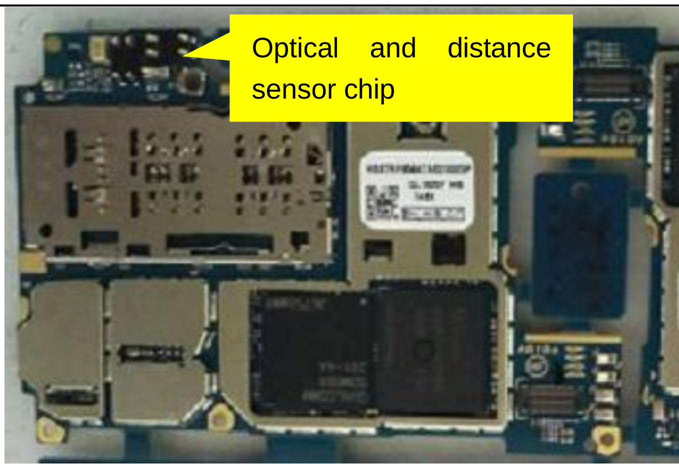
1 Check the MB, connector FPC, connector pins and their connections. Re-assemble or replace them before rebooting and testing.
2 Check the LCD and CTP connector (J2001) for dirt, PIN short bridge, cold solder joint, PIN collapse, etc.
3 Measure (VREG_L6_1P8) supply voltage and check whether or not it is normal. Measure each signal pin impedance and check whether or not it is normal;
4 Check the CPU soldering status with an X-ray, add-solder or re-solder it before testing it again;
5 Re-solder or replace the CPU (U801);

6. Optical/distance sensor error

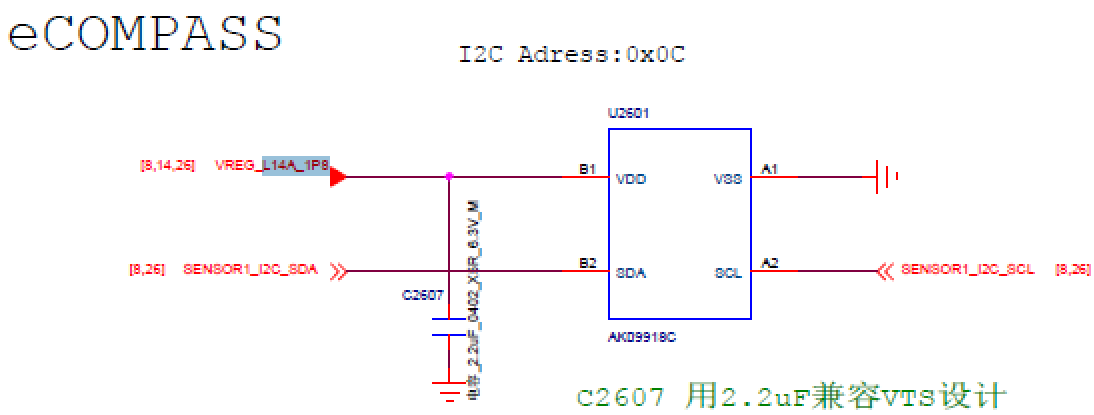


Solution:

1 Check optical/distance sensor for damages or soldering errors;
2 Measure the voltage VREG_L3B_2P85 across R2601 and check whether it is normal.
3 Measure the voltage VREG_L14A_1P8 across R2602 and check whether or not it is normal.
4 Measure each signal pin impedance of the optical and distance sensor, and check whether or not it is normal
5 Replace the optical and distance sensor chip.
6 Add-solder or re-solder the SDM636, or replace it.

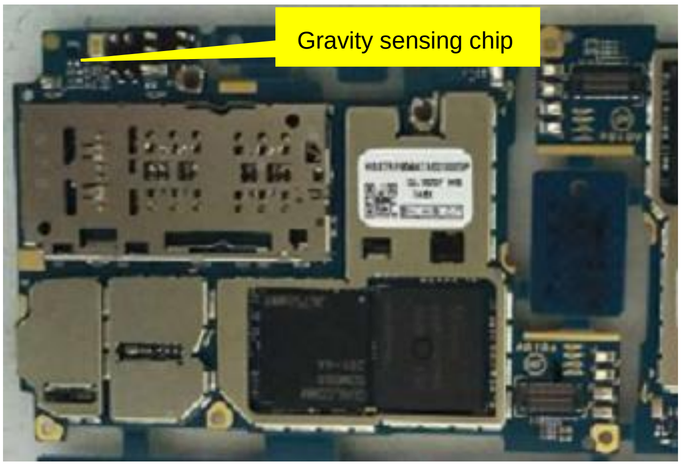


7. Gravity sensor

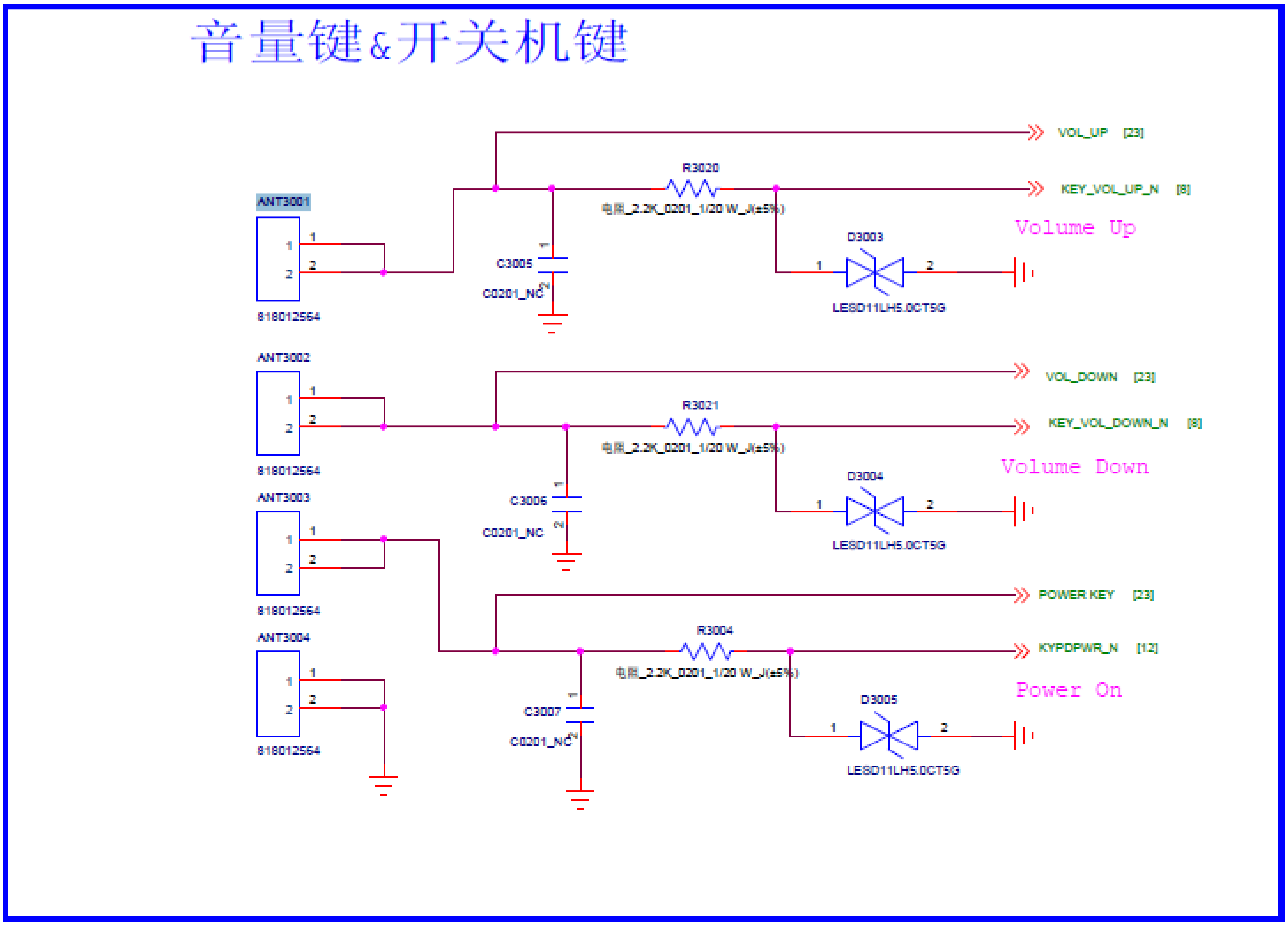


Solution:

1 Check the sensor module circuit components for soldering errors or damages;
2 Check whether or not it is functionless or has a function error;
3 Examine the peripheral components and access components;
4 Re-solder or replace the sensor U2601;
5 Re-solder or replace the CPU;



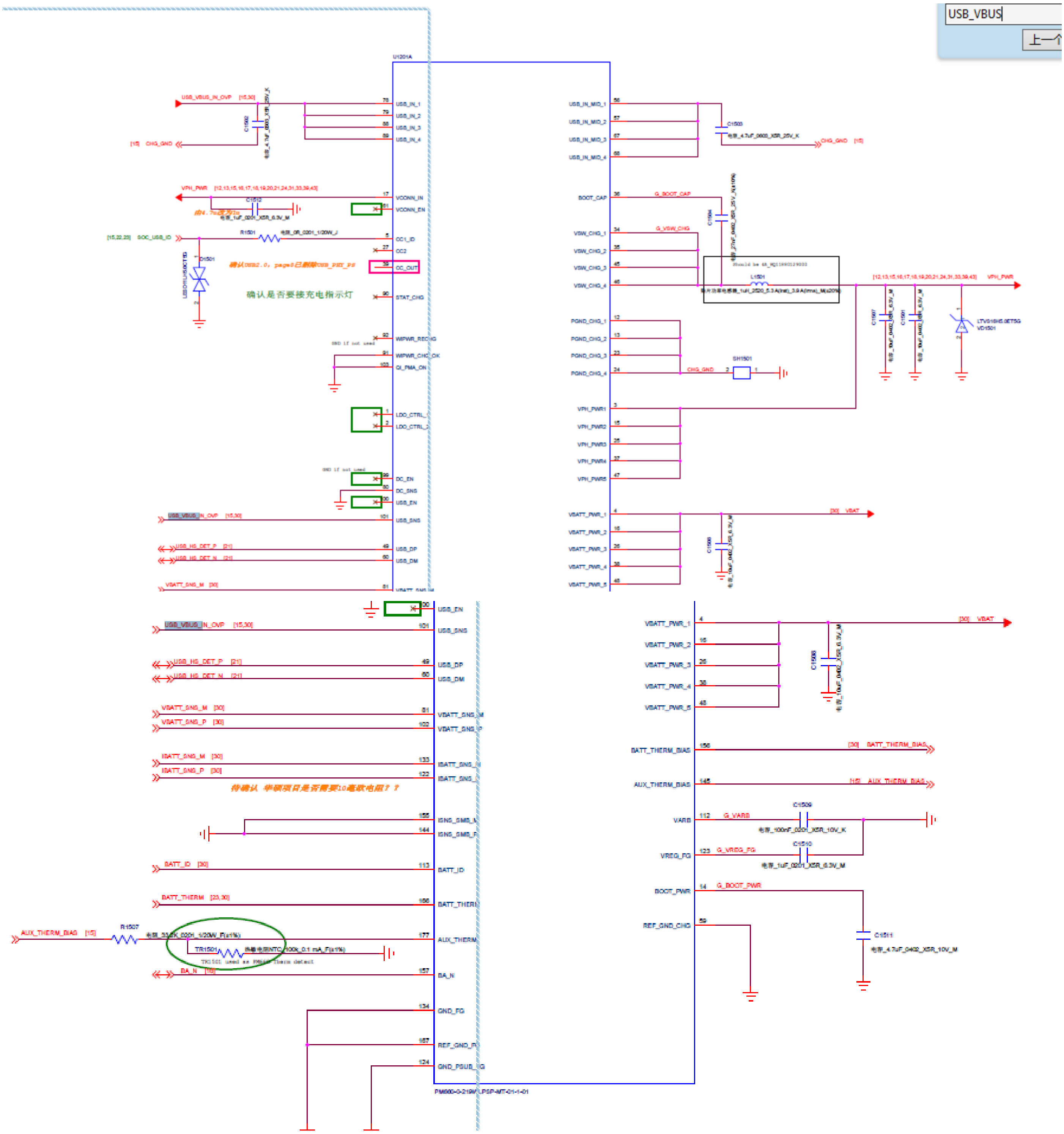
8. Key failure



Solution:

1. Check whether or not the key is dirty or obscured by foreign objects.
2. Check Volume UP/Volume down/Power-On voltage signal validity;
3. Measure each key signal impedance, check whether or not it is normal.
4. Check components on the key circuit for damages or soldering errors;
5. Re-solder or replace the PM660L and SDM636.

9. Not charging



Solution:

1 Check whether or not the SB board or FPC which connects to the MB is normal. Replace the parts and determine the failure causes.

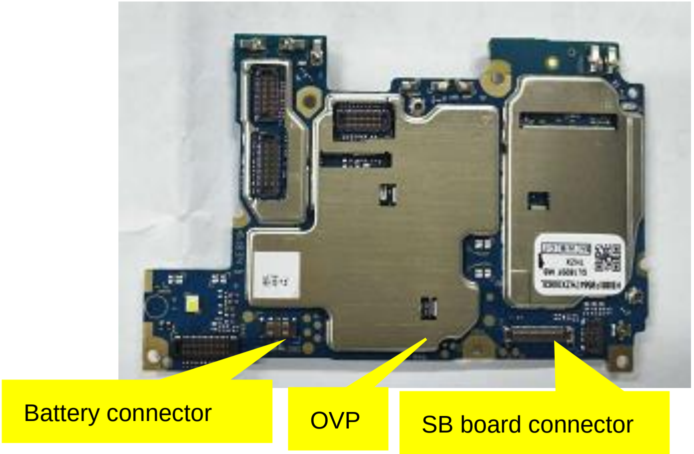
2 Check whether or not the impedance of each pin of battery connector is normal and compare it with that of a normal MB. Check circuit components of the pin with abnormal impedance.

3 Replace the battery before testing.

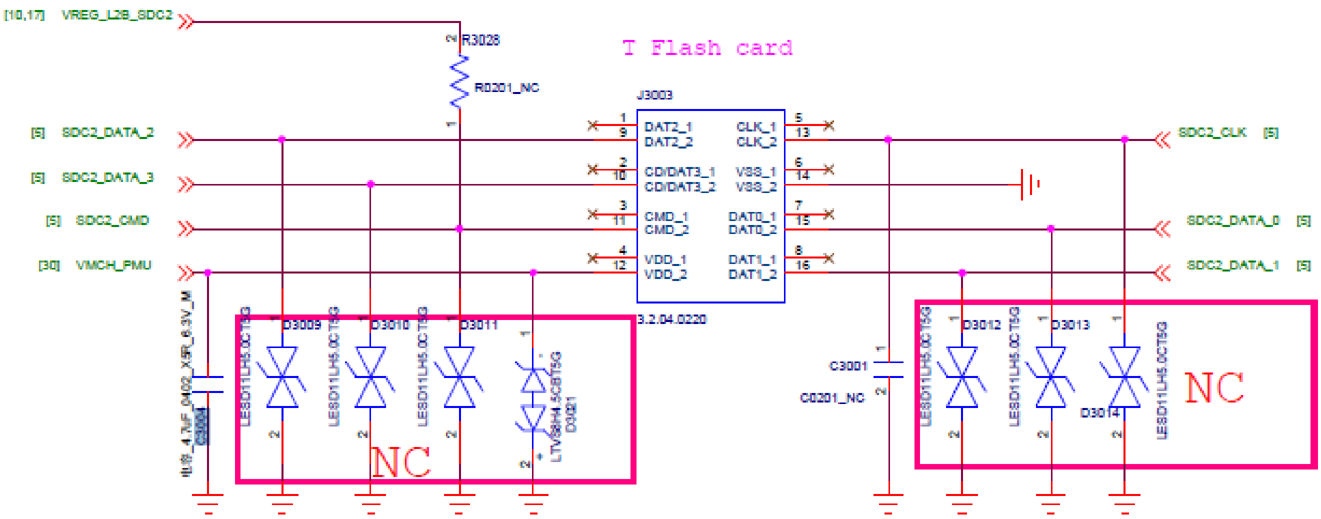
4 Check whether or not the voltage USB_VBUS_IN_OVP at C3012 is normal. If not, check whether or not the soldering of the OVP component U3001 is normal and whether or not its surrounding components are normal.

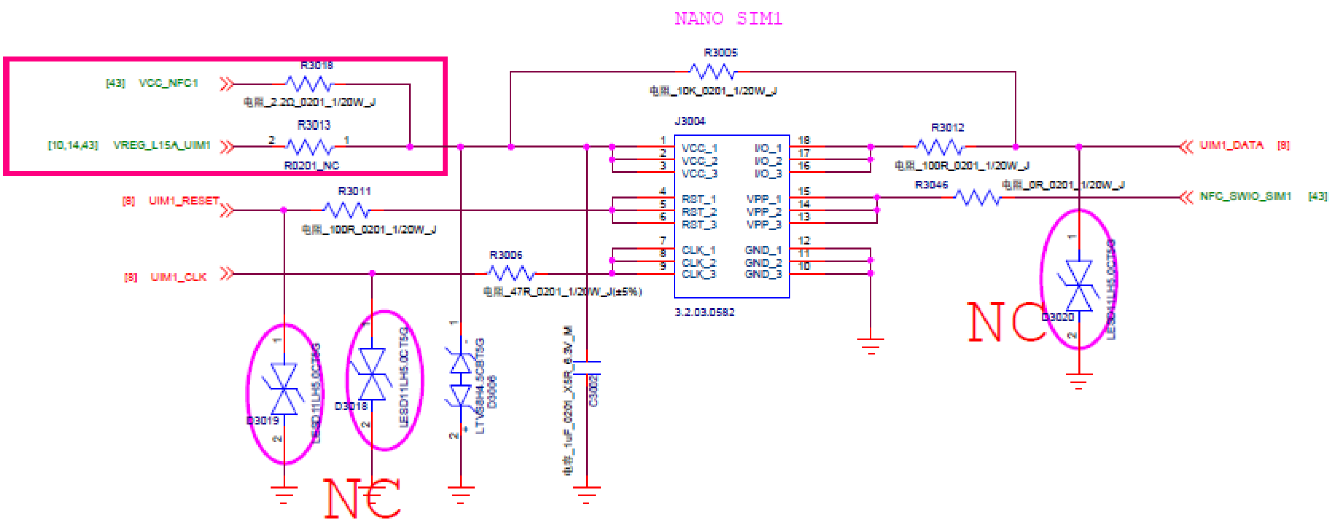
5 Insert a charger and check whether or not the voltage signal on the C1508 is normal. If there is no

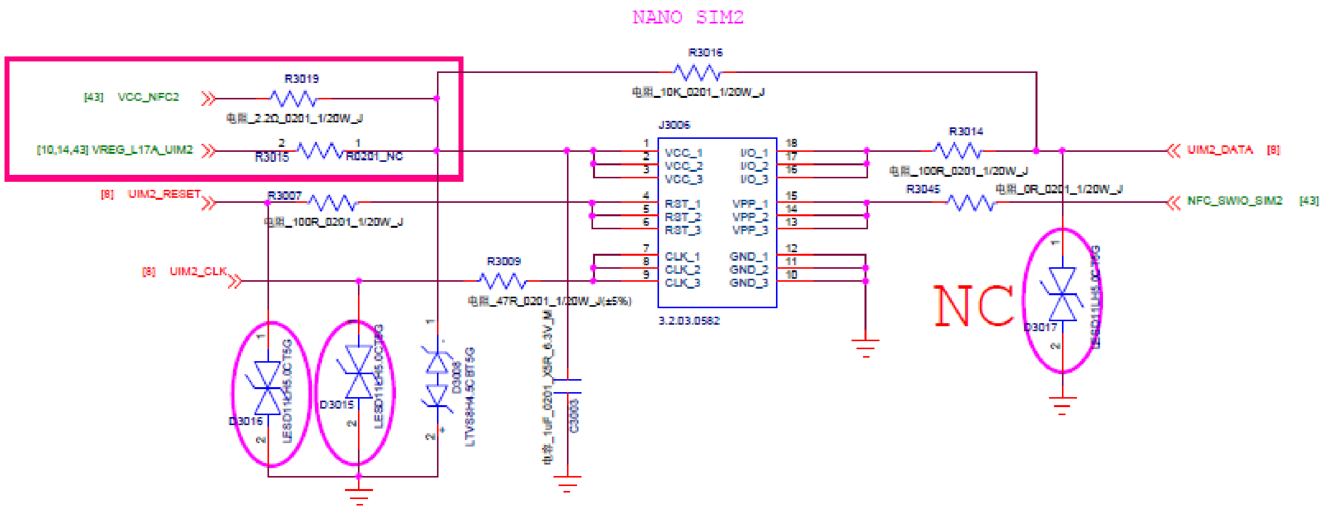
charging voltage, check whether or not the PM660 (U1201) or its soldering is normal.



10. SIM card or SD card identification failure







Solution:

1 Visually check the metal cover of the SIM tray for warping or deformation, the detecting pin on the tail for contact errors and the push rod for proper operations with no resistance.

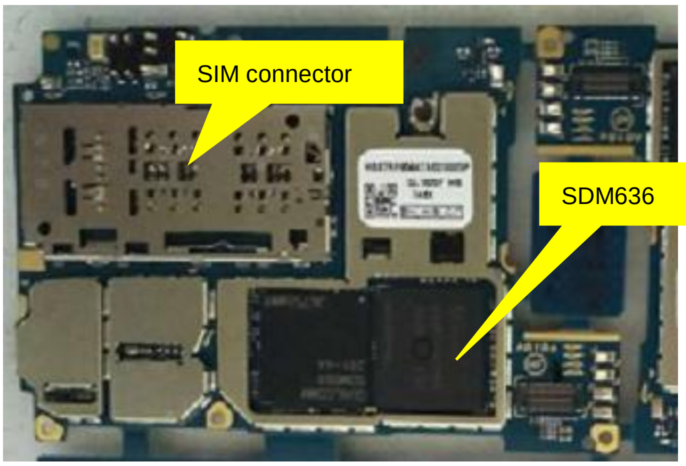
2 Check the SIM card and SD card connector pins in the metal cover for soldering errors and the metal pins in contact with the cards for dirt or collapse.

3 Measure the impedance of each functional pin with a multi-meter and check whether or not it is normal (compared with a normal MB).

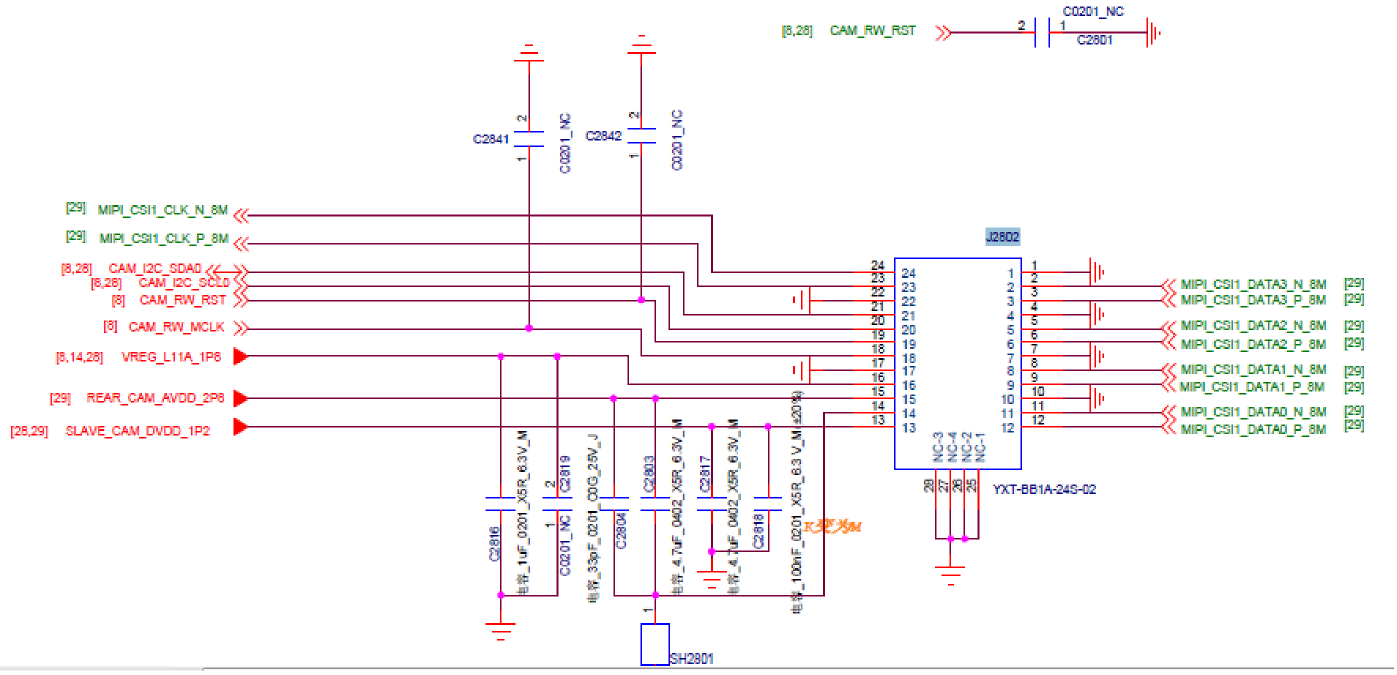
4 Measure the power voltage signals VREG_L2B_SDC2/VMCH_PMU(C3004)/VREG_L17A_UIM2(R3015)/ VREG_L15A_UIM1 (R3013) and check whether or not they are normal

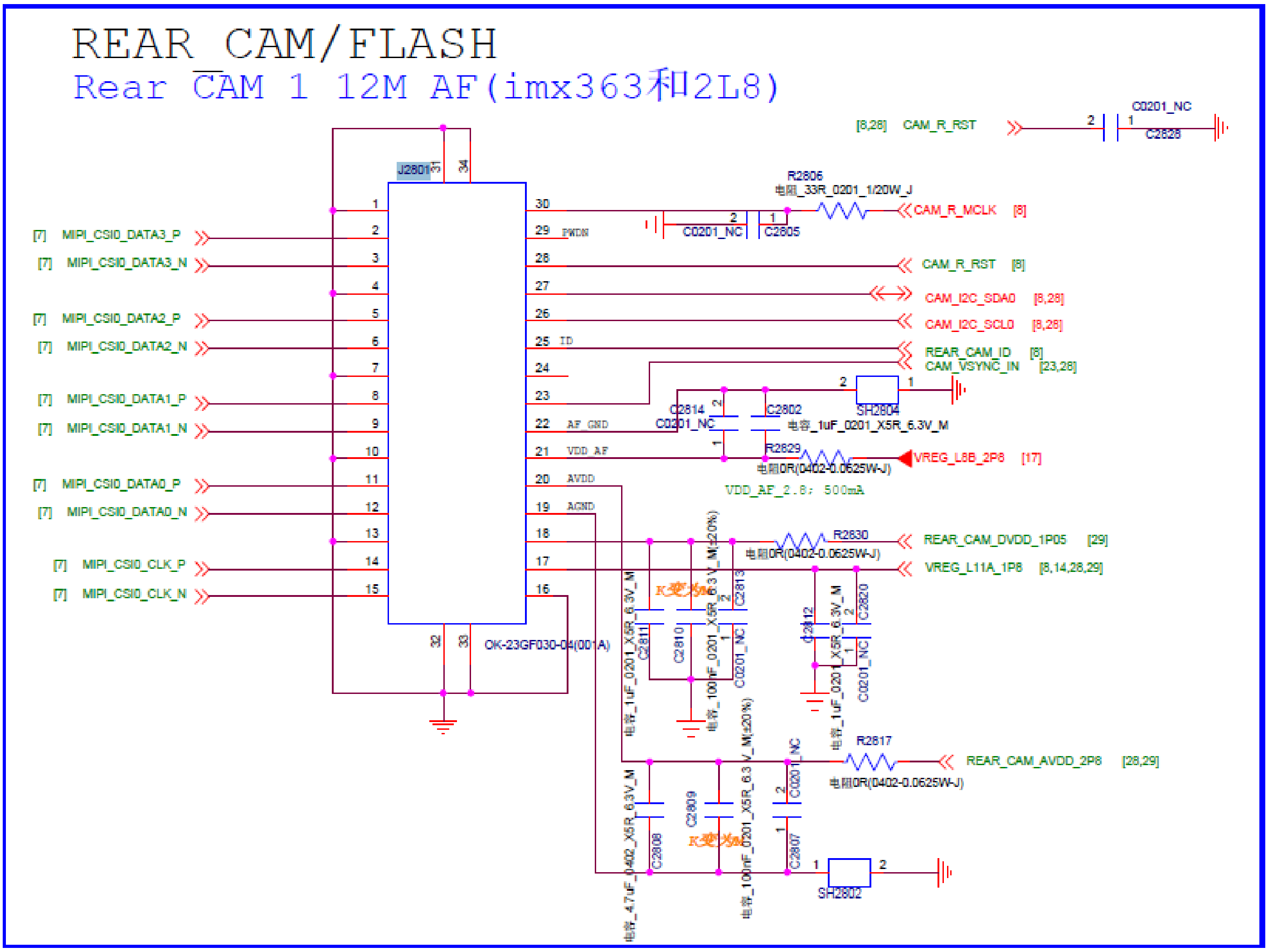
5 Check the SIM and SD card function-related components for soldering errors.

6 Re-solder or replace the SDM636.



11. Main/auxiliary camera function failure





Solution:

1 Check the camera installation for any abnormalities, re-install it or replace it before testing again;

2 Check the connector of the main camera J2801 and the connector of the auxiliary camera J2802 for damages or soldering errors. Re-solder or replace the connectors.

3 Check whether or not the voltage signals VREG_L8B_2P8, REAR_CAM_DVDD_1P2,

VREG_L11A_1P8, REAR_CAM_AVDD_2P8 are normal.

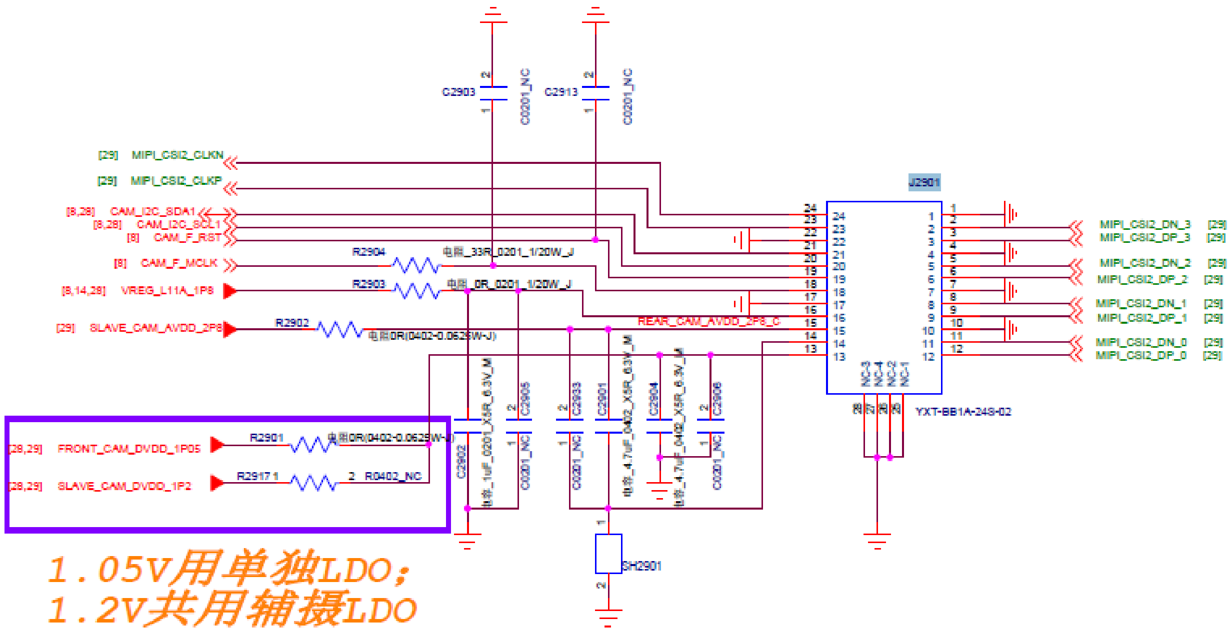
4 Check the EMI filter components T2806-T2810 for soldering errors.

5 Measure the impedance of each pin and check whether or not it is normal. Check the circuit components and chips at the abnormal impedance point for soldering errors or body damages.

6 Check the LDO chip module components U2902/U2904/U2905 for soldering errors.

7 Re-solder or replace the SDM636.

12. Front camera function failure



Solution:

1 Check the camera installation for any abnormalities, re-install it or replace it before testing again;

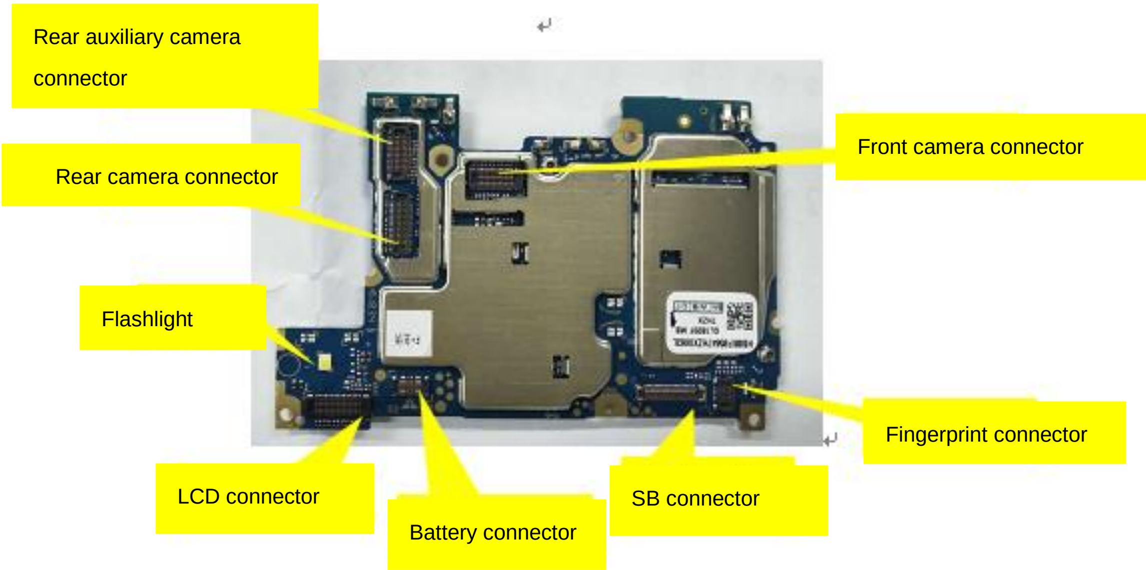
2 Check the auxiliary camera J2901 connector for damages or soldering errors. Re-solder or replace the connector.

3 Check whether or not the voltages VREG_L11A_1P8, REAR_CAM_AVDD_2P8, FRONT_CAM_DVDD_1P2 are abnormal. If so, check whether or not it caused by the PMIC output error or by a short or broken circuit in the peripheral circuits.

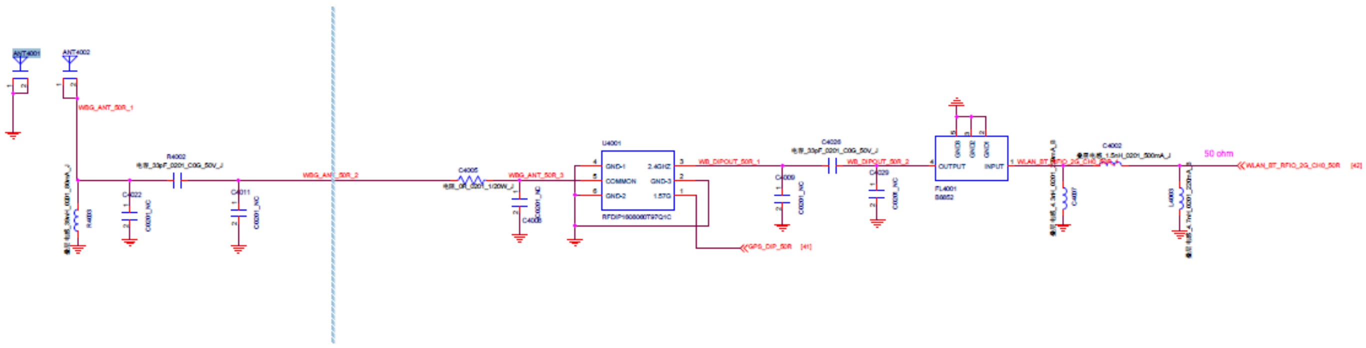
4 Check the EMI filter resistor for damages or soldering errors.

5 Check whether or not the LDO chip U2906 or its soldering is normal;

6 Add-solder / re-solder the SDM636



13. Motor error

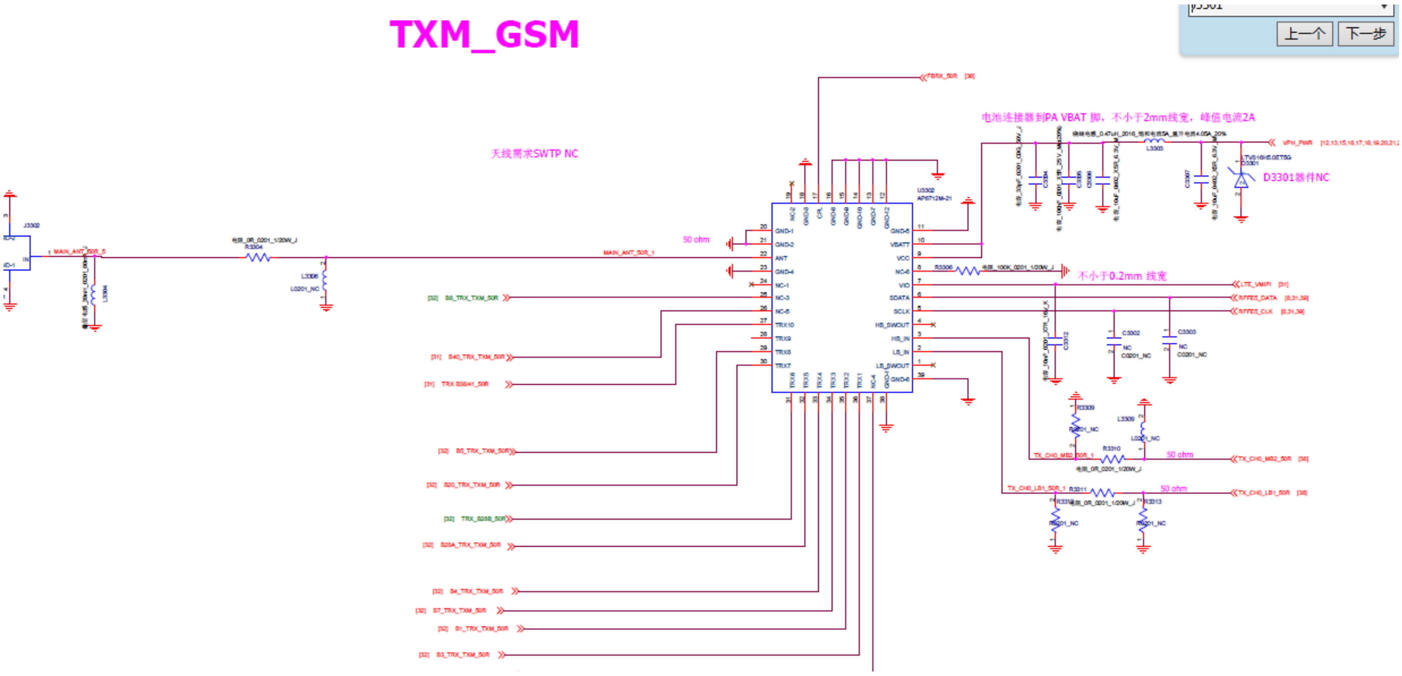


Solution:

1 Check whether or not the resistance and soldering of the R4002 and R4003 are normal;
2 Check the ANT4001/ANT4002 contact point for foreign objects. Check whether or not the impedance value is normal;
3 Replace the motor;
4 Re-solder or replace the PM660 (U1201).

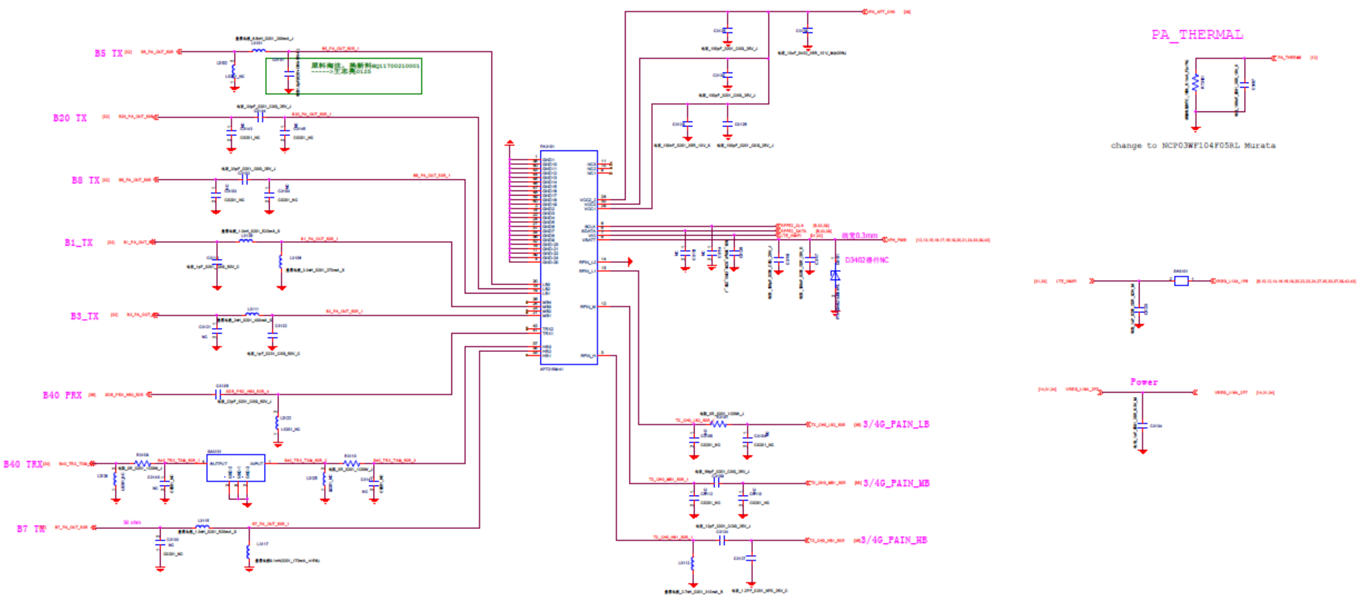
14. GSM network failure

Solution:



1 Check the PCBA board's antenna for soldering errors, its contact with phone casing antenna and body damages;
2 Check the connection of the RF connector and circuit components, inductors, resistors for soldering errors after collision;
3 Check the connectors J3301, J3302 and relevant components for soldering errors and body damages.
4 Check whether or not PA U3302 and its soldering are normal.
5 Re-calibrate the MB RF indicators. Find the circuit components with corresponding the frequency according to the calibration results.
6 Add-solder or replace the U3601 RF transceiver chip;
7 Add-solder or replace the SDM636.

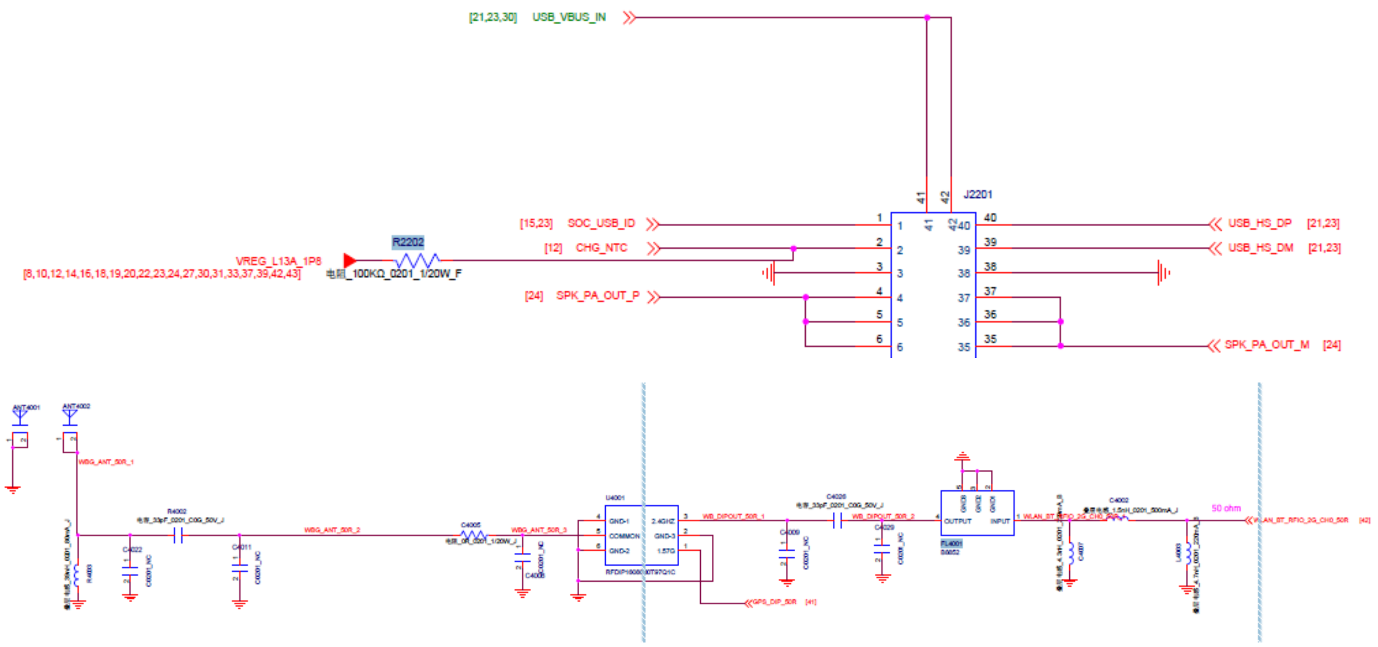
15. Network failure for the LTE frequency bands



Solution:

1 Check the primary and diversity test stands for false solder joints;
2 Re-calibrate the phone, pinpoint cause of error, determine the receiving or transmitting errors according to the bands with errors, troubleshoot the relevant circuit components;
3 Search matching circuits according to the corresponding frequency bands in the schematic diagram. Search components for the corresponding frequency bands to analyze according to the failure code in the calibration and comprehensive tests.
4 Check the crystal module circuit for soldering errors. The frequency-related failures can be checked first.
5 Re-solder or replace the PM660
6 Re-solder or replace the SDM636

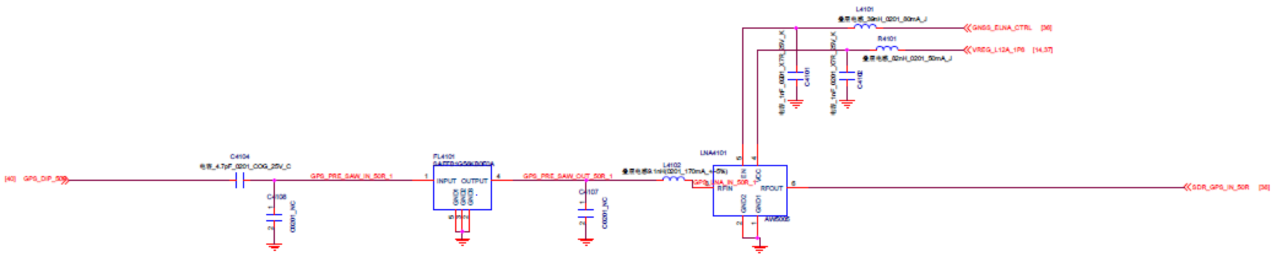
16. WIFI/Bluetooth/FM



Solution:

1 Check the MB's WIFI/BT antenna channel components or their soldering for damages;
2 Check the RF test stand for false solder joints;
3 The FM antenna module is on the SB board. Check whether or not the KB board or FPC is normal. Re-assemble or replace it before testing.
4 Check the filter FL4001, U4001 for soldering off-set or body damages.
5 Check the resistors, capacitors and inductors in the circuit for missing, tombstone, off-set and other soldering errors.
6 Check the WCN chip U4201 and its peripheral matching circuit components for soldering errors. Re-solder or replace the WCN chip before testing
7 Add-solder or re-solder the SDM636, or replace it before testing.

17. GPS failure

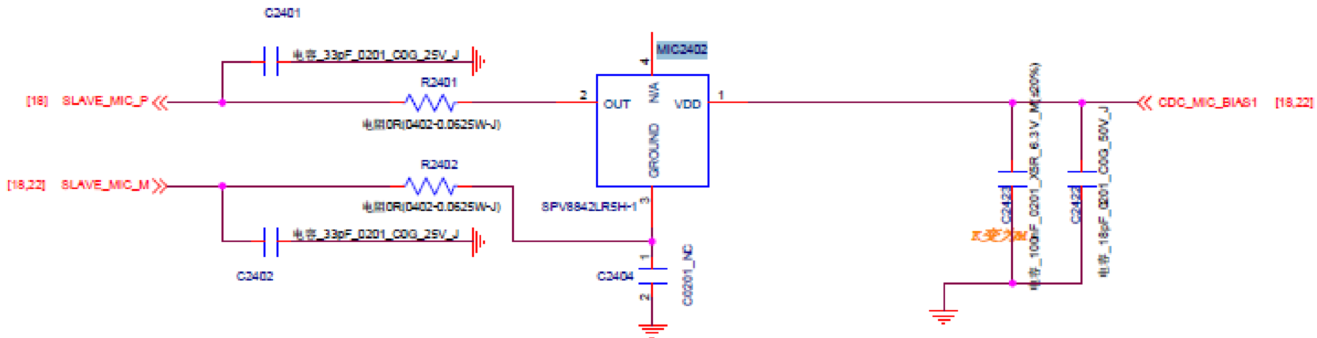


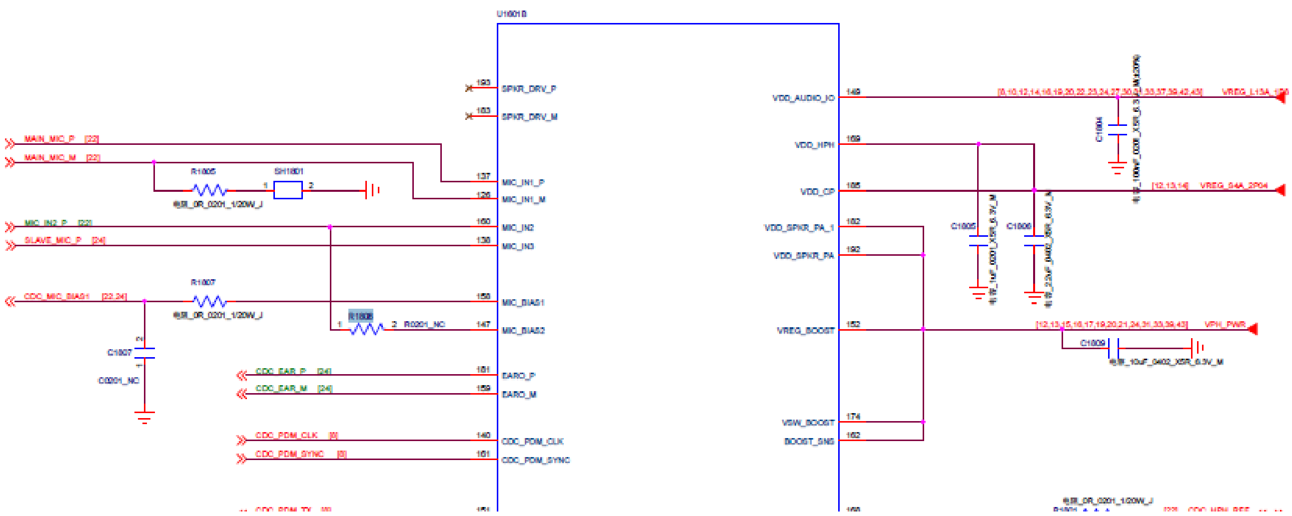
Solution:

1 Check the GPS channel components for soldering or body errors;
2 Check the RF test stand for false solder joints;
3 Check the filter FL4101, FL4102 for soldering or body errors.
4 Check the LNA chip LNA4101 for soldering or body errors.
5 The GPS antenna circuit is shared with the WIFI and BT modules. Check whether or not the WIFI and BT functions fail at the same time.
6 Add-solder or re-solder the U3601 RF chip or replace it before testing.
7 Add-solder or re-solder the PM660 or replace it before testing.
8 Add-solder or re-solder the SDM636, or replace it before testing.

18. MIC silent/low volume

Slave MIC





Solution:

1 Visually check the auxiliary MIC, single MIC2402 and relevant components for soldering errors or body damages.

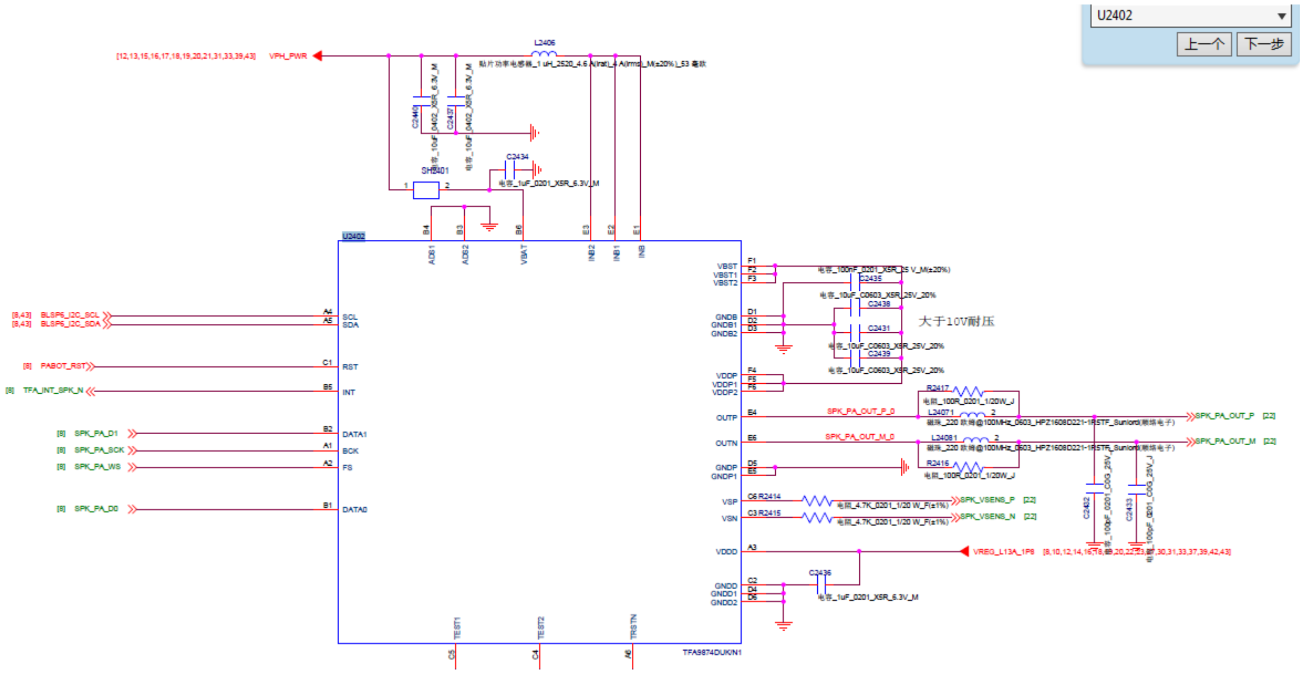
2 The main MIC is on the auxiliary board. Replace the KB board and FPC to test and check whether or not the source of fault is in the MB circuit.

3 Check soldering ring tray of the MIC sound hole with an X-ray for broken ring, which easily leads to test failures for machine audio indexes.

4 Observe under an X-ray whether or not board ashes, solder paste, flux and other foreign objects have entered the sound hole.

5 Add-solder or re-solder the PM660L or replace it before testing.

19. Speaker silent



Solution:

1 The SPK module is on the SB board. Re-assemble the KB board or FPC for testing.

2 Replace the FPC and KB board to determine whether or not the source of the fault is on the MB.

3 Check the MB’s KB connector, measure the SPK signal and supply voltage signal, and check whether or not they are normal.

4 Add-solder or replace the Audio PA U2402.